Vehicle Anti-theft Tracking System Based on Internet of Things

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***Abstract*—As the amount of urban vehicle grows rapidly, vehicle theft has become a shared concern for all citizens. However, present anti-theft systems lack the tracking and monitoring function. The vehicle anti-theft tracking system based on Internet of things is designed in this article, which can provide all-round active service for the owners. This system is controlled by an RFID module to switch on and off. When the car is stolen, the vibration sensors and pyroelectric infrared sensors mounted inside the vehicle are triggered, and GSM module will send the location information obtained by GPS module to the owner's mobile phone, thus owners can check the position of the vehicle with an android software developed by our team. This system uses android mobile phones as mobile terminal, which is more convenient and flexible than other kinds of like products since the owner can check and track the position of the car immediately with android mobile phone application once the car is stolen.**

Keywords—Vehicle anti-theft system; RFID; GPS; GSM; Internet of things

I. INTRODUCTION

As the number of urban vehicles grows rapidly with the development of the economy, people are getting more concerned about vehicle theft prevention, which creates broader market prospects for vehicle anti-theft products.

Various vehicle anti-theft devices have been developed lately, however the result is still disappointing since all kinds of devices have its drawbacks. Domestic and overseas vehicle anti-theft products are technologically classified into three categories: mechanical lock devices, car alarm system, and vehicle tracking/recovery systems, mainly aiming at preventing cars to be broken in and driven away. The most commonly used mechanical lock device is steering wheel lock, which is relatively cheap but inconvenient to use and may be easily disarmed by skilled thieves. Car alarm devices are very popular these days [1]. However, statistics show that 96% of the public are not aware when they hear an alarm because of the constant false alarms which also disturbs live of residents around. These car alarm systems do not cover large areas; the area is just less than 100m [2]. Once the car is stolen, the owner and the police can not track the position of it.

The commonly used vehicle tracking/recovery systems are based on radio signals such as the LoJack tracking system, the ProScout GPS Vehicle Tracking System, the TravelEyes2 Vehicle Tracking System and so on. After a vehicle has been stolen, the owner can report the problem to the police or the GPS tracking office. The wireless transmitter or the GPS device in the car will send wireless signals which can be picked up by the tracking device. The wireless signals can be used to pinpoint the location and lead police to rapid recovery. However, these systems have high cost and often come with a monthly monitoring fee [1].

To address the limitations of existing vehicle tracking/alarming systems, we propose an anti-theft system mainly based on GPS and GSM system, designed and realized by integration of traditional anti-theft alarm、GPS、GSM SMS system and mobile phone android applications. The owner will receive a short message sent by GSM module as soon as the car is stolen, then he can use android mobile phone application to track down the car. Compared with traditional vehicle anti-theft systems which require customers to install software and track the position of the car on computers, our system use android mobile phone as client terminal, providing customers with more immediate, convenient and cheaper service. Additionally, since we use free Google Map for location, which means that clients only have to pay for WAP data traffic and SMS costs. Among them, SMS module costs less than ￥5. 00 monthly if the owner has ordered text message packages launched by mobile telecom carriers. For further development, manufacturers cooperating with mobile telecom carriers can allow customers to send and receive messages at very low cost.

II. SYSTEM STRUCTURE

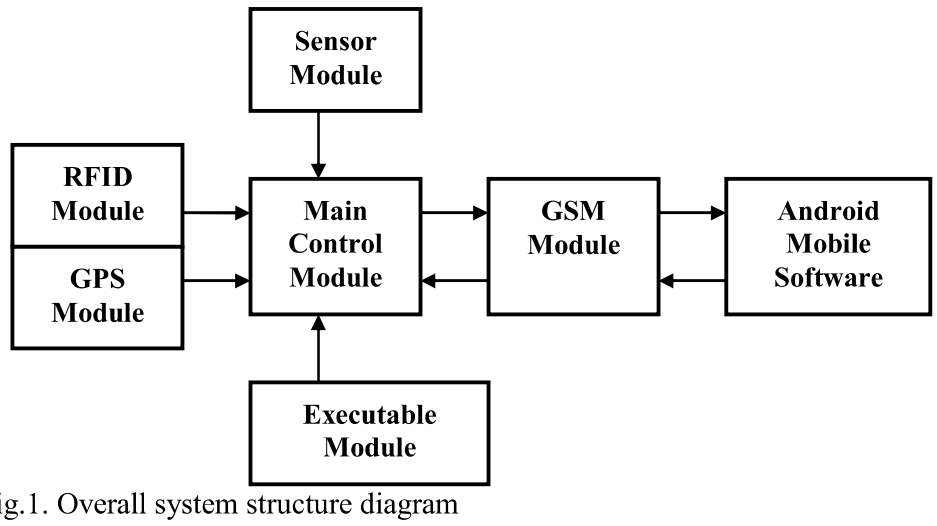

Fig.1. Overall system structure diagram

A. RFID Module

We use an RFID device embedded inside the car to control the power switch, which is less easy to be damaged. The RFID system is composed of two basic components: tags or labels attached to the objects to be identified and readers to send signals to the tag and read its response. RFID tags can be either passive, active or battery assisted passive. We use active tags into consideration that the tag carried by the owner needs to be inducted at any position in the car. When the owner stays inside the car, the reader receives responses returned by the tag and transmits a signal control relay to switch off the power supply unit. Otherwise, when the owner leaves the car, bringing the tag outside of the interrogation zone of the reader, the device starts working.

B. GPS and GSM integration module

After the power supply unit switch on, GPS (Global Positioning System) stores the latitude and longitude information it collects in E^2PROM chip. When thieves get into the car, the vibration sensors mounted in the doors will be triggered. We also use pyroelectric infrared sensors to make sure it's caused by human factors. When both sensors are triggered, GSM (Global System for Mobile Communications) module will send GPS information in the E^2PROM chip in certain form to the specified mobile phone of the owner in a certain time interval.

1) GPS Module:

Serial output format of GPS positioning information uses the NMEA-0183 communication standard developed by the U.S.National Marine Electronics Association[3].Output data using ASCII code contains the latitude, longitude, altitude, speed, data, time, heading and satellite status information. There are six kinds of commonly used sentences, including GGA、GLL、GSA、GSV、RMC and VTG. Output logging statement and transmission time interval are set depending on different application demands. As for our device, we only need latitude and longitude data, so we choose GGA、VTG logging statement which is transmitted every 1s.These settings are only valid during current working status

In this device, required longitude and latitude data need to be extracted out from information received GGA statements. After the system successfully initialized, six integer numbers of degrees, minutes, seconds of longitude and latitude are stored in E^2PROM and updated correspondingly when the position of the device changes in real time. A significant benefit of GPS module working independently is that the accuracy of the data received each time is guaranteed.

2) GSM Module

We used Siemens GSM module in this device [4], which supports dialing and sending Chinese messages. It works between the frequency band of 1800MHz and 1900MHz, and power supply ranging from DC 3.3V to 4.8V.The GSM module transmits speech signal and a data signal with power consumption of 2W at EGSM900 (Class 4) and 1W at GSM1800 (Class 1), connects with SIM card reader and antenna via interface connector and antenna connector. It also supports Text and PDU format of SMS [5].

C. Main Control Module Design

There are two single chip microcomputers (SCMs) working independently. One acquires the GPS data and stores them in E^2PROM chip. When vibration sensors and pyroelectric infrared sensors are triggered at the same time, another SCM will gather the longitude and latitude information in EEPROM chip and transmit via GSM module.

III. SOFTWARE DESIGN

A. Overall Working Flow

The device switch is controlled by RFID module. An active RFID tag is attached to the car key and the reader is fixed inside the car. The device doesn't work when the owner stays in the car, and starts working when the owner leaves the car. When the car is intruded anomaly by criminals, the vibration sensors will be triggered. To ensure it's caused by anthropic factors, we also install pyroelectric infrared sensors. GSM module will send information collected by the GPS module via SMS to the owner's mobile phone every 10 seconds. The owner can open the android application after receiving messages, click on "Vehicle Position" to check the position of the vehicle. When the owner finds the car stolen, he can communicate with the GSM module on the car via Android software. Android software can send short message to GSM module, then GSM reports the information to master control module, which is able to forcibly lock and unlock the car.

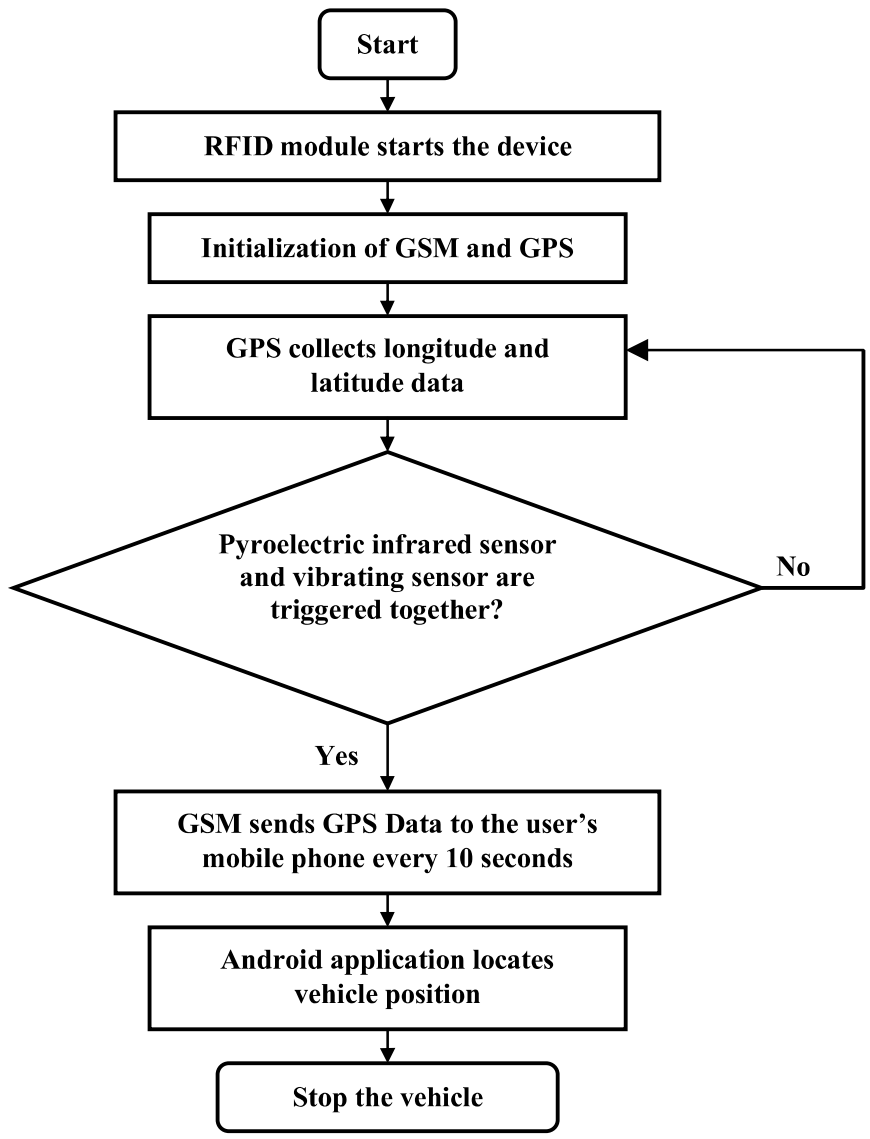

Fig.2. Working flow diagram

B. *Android System*

Android is a Linux-based, open source operating system developed by Google. It is designed primarily for smart phones [6] and tablet computers, with operating system, middleware and applications. Written by source code, Android can be transplanted to different hardware platforms.

The first layer of Android is realized by language C, the second layer is realized by C and C++, and the third layer is written in JAVA. Between the first and the second layer is the division of kernel community and users' space. The first layer runs on the kernel community while other layers run on the users' space. And between the second and the third floor is the interface of native code layer and JAVA code layer. The interface of Android system and Application programming interfaces (APIs) lies between the third and the fourth layer. Contents of layers below the third layer are invisible, so only the API takes into consideration when programming. So Android application development should mainly focus on the interface between the third and fourth layers.

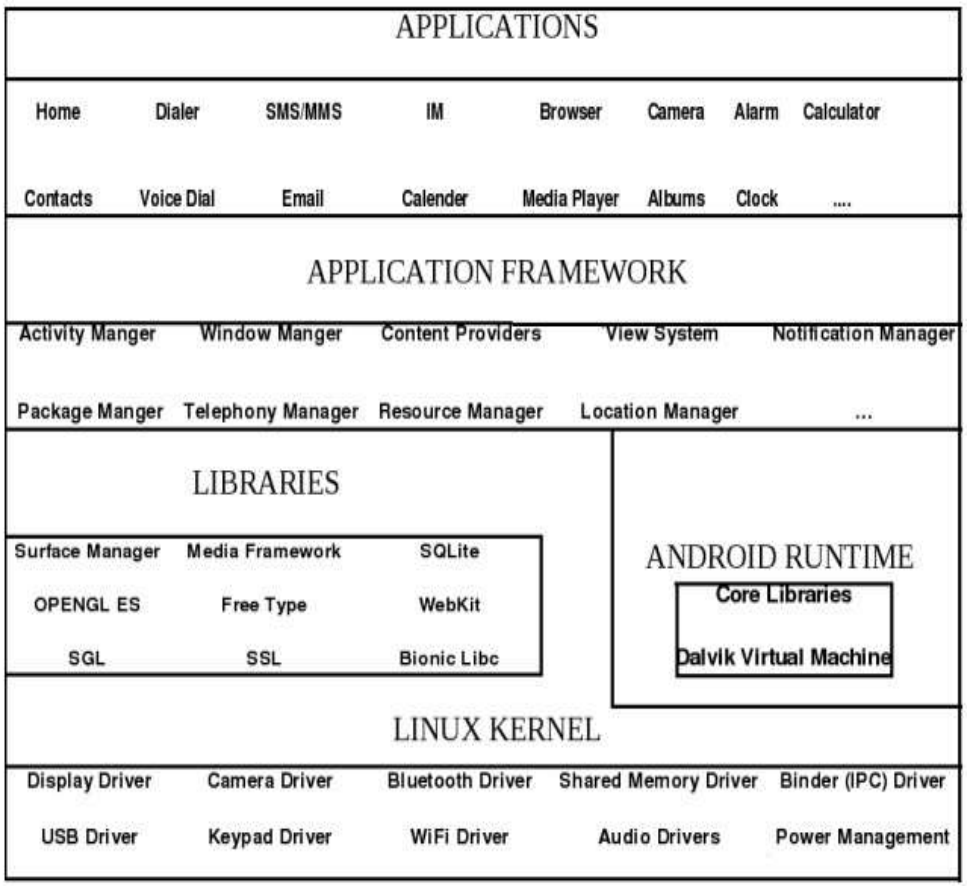

Fig.3. Android system structure

C. *Get the Short Message*

The database path of Android mobile phone system short message is:

"/data/data/com.android.providers.telephony/dababases/mmssms.db"

Generally, the operation of SMS database is mainly related to three tables: Canonical_addresses、Threads、SMS.

Through three connectedly tables, we are access to all short messages、conversation lists and contacts, short message lists. As for this software, we only need the longitude and latitude information, so only SMS table is required.

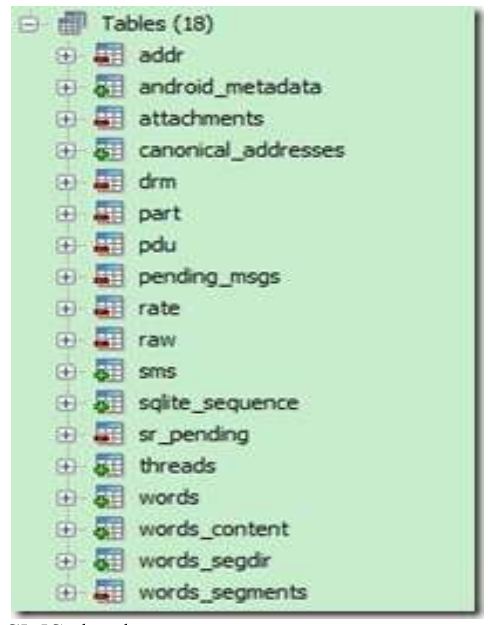

Fig.4. The android SMS database

SMS table can be queried by user-defined URL, for example, content: *//sms/*. We find during debugging that it becomes very slow querying all messages at a time when the application processes many messages. To solve this problem, only the content of the latest received message is extracted during each operation, so that we can maximize efficiency and achieve "locate by one click".

D. *Google Map Locating*

Ever since mobile phone started incorporating devices that made them aware of their geographic locations, developers have foreseen a new era of location-base applications. Location awareness improves many applications and makes possible totally new applications. Google's Android offers powerful features that make the development of these applications very easy.

Before using Google Map service, we have to apply for an android Map API Key by registering at Google official website. A certain decoding algorithm matching with the fixed string format of the message content is also needed to extract the longitude and latitude information in string.

E. *AndroidMainifest.xml File: Permissions*

Every Android application includes an *AndroidMainifest.xml* file [7] in the root directory. This file presents essential information about the particular application to the Android System. Some of the elements included in the structure of the AndroidMainifest.xml are the <users-perssion> elements. The <users-perssion> element requests a permission that the application requires in order to operate correctly. Android botnets commonly use the following permissions.

- <uses-permission android:name="android.permission.READ_SMS" />
- <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
- <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>

- <uses-permission android:name="android.permission.ACCESS_MOCK_LOCATION"/>
- <uses-permission android:name="android.permission.INTERNET"/>

The *AndroidMainifest.xml* file provides valuable information to the user about a particular application and contains identifiable characteristics of an Android botnet [7].

F. SMS Control Protocol

When the car is stolen, the owner can use android application to send short message to GSM module of the device on the car [8] to conduct the available operation of locking the car, unlocking the car and cutting of gasoline according the content of the short message. We also set a series of protocol for the communication between owner and GSM module. The owner's mobile phone will receive a short message in fixed format of *password#dddddd#*, among which 'password' is a fixed code that every user hold and 'dddddd' is a command parameter standing respectively for different operation to the car. For example, it is prescribed that 000001 locks the car, 000002 unlocks the car and 000003 cuts off gasoline of the car. In consideration of personal security, the owner should contact the police and provide position information on the stolen car to solve the problem.

IV. TESTING RESULT

When the car is stolen, the specified mobile phone of the owner receives messages as shown in Fig.5, which contains longitude and latitude information. The owner opens the android application software, clicks on "Vehicle Position". Then the software will fetch the position information directly from the message and transform it to coordinate displayed on the map.

Fig.5. Short messages received by mobile phone

Getting the integer value of longitude and latitude, we can use Google Map API. Firstly, we construct a GeoPoint to indicate the longitude and latitude. Then we locate the map to the GeoPoint via animateTo method. Plus, some icons used as orientation information are needed, which are realized by Overlay. The longitude and latitude datum are converted into coordinates via the method provided by Map API so that they can be displayed on the cell phone screen.

Fig.6. Present location of the Car on Google Map

V. CONCLUSION

The article has described a vehicle anti-theft monitoring system based on GPS technology and GSM technology, cooperating with android software positioning, which contains a communication platform with remote monitoring function. The owner can simply use mobile phone to locate and monitor the car in real time, which provides the technical foundation for the development of wireless network vehicle anti-theft system.

For future works, we can extend the use of this system to the car dispatching management field, which has vast potential for future development.

ACKNOWLEDGMENT

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